IN5450/9450
Mandatory Exercise 1

Andreas Austeng

February, 2026

Distribution date	February 17, 2026
Deadline	March 9, 2026
Submit solution in	https://devilry.ifi.uio.no
Version	1.0, February 17, 2026

- In this exercise we will study diffraction investigating the response calculated using the Rayleigh-Sommerfelt (R-I)integration formula and the array pattern (the aperture smoothing function) of several arrays.
- Problem 1 – 2, 4 – 9, and 12 – 13 are mandatory for IN5450 and IN9450 students.
- All other problems are mandatory for IN9450 students, but voluntary for IN5450 students.
- MATLAB code for R-I and for computing and analyzing the array pattern are provided. It can be reused in most of the exercises below.
- You may chose whatever programming language you prefer. The text, though, assumes you are using MATLAB.

Rayleig-Sommerfeld integration formula

Diffraction and the Rayleigh-Sommerfeld integration formula was discussed in the first part of the course. A text describing it can be found in the course curriculum written by Sverre Holm, Ch 7. The R-L Diffraction formula describes the field $s(\vec{x})$ at a given position $\vec{x}$ occurring as a result of a field $s(\vec{x}_h)$ from an aperture A (described by $\vec{x}_h$):

$$s(\vec{x}) = \frac{1}{\imath\lambda} \iint_A s(\vec{x}_h) \frac{\exp\{\imath k r\}}{r} \, dA. \tag{1}$$

Diffraction from a line aperture

In the first part of this exercise we will study the diffraction field from a line aperture. The aperture is located along the x-axis and centered in origin. We study the field along the z-axis. The provided MATLAB script *DiffLinAp.m* propagates a uniform pressure distribution of a line aperture with length $D = 20\lambda$, excitation frequency $f_x = 3$ MHz, and positioned at $z_0 = 0$ to a chosen range $z = 5$ cm and plots the result using the MATLAB function *R_S_LinAperture.m*.

1. Run the script *DiffLinAp.m*. Do the plots look as expected? Explain what you observe and why. (Hint: If you for some reason are unsure about what to answer, continue on the following tasks and return to this later).

By changing the observation points defined in *DiffLinAp.m* it is possible to calculate the diffraction response at other locations. If one wants to image the response in a 2D plane, one could change the code to something like

```
> r = 0.08;
> z = 5*dz:2*dz:r;
> x = -r:2*dz:r;

> [X,Z] = meshgrid(x,z);
```

2. For the same setup as above, modify the *DiffLinAp*-script to predict the diffraction field for the scenario described below:
   - Make a 2D image of the diffracted field for $x \in [-8, 8]$ cm and $z \in [5\lambda, 8\text{cm}]$.
   - Plot the pressure along the z-axis (broad side the array) for $x = 0$ (i.e. a 1D plot, along the z-axis, and starting at the origin).
   - Plot the response along a half circle at the radius given by the the near-field farfield limit. Describe how you defined this limit. (Hint: The near-field farfield limit for the line array does deviate from the one of a piston).
   - Plot the response along a half circle at the radius much larger than the near-field farfield limit.

   Report all figures. Remember axes/labes/titles/etc. Discuss your findings!

3. Implement weighting of the aperture. Investigate how weighting affects the near-field far-field limit of the aperture.
   Report figures. Remember axes/labes/titles/etc. Discuss and explain your findings!

Array Pattern

The first ting to do:
You may write your own MATLAB function for computing the array pattern (aperture smoothing function) of an M-element array with the elements located on the x-axis. Assume initially that each element is omni-directional. The weighing (a vector, one value for each element) should be an input variable together with a vector of element positions, and a vector of values of k_x (or $\sin(\text{angle})$ or angle) where the response is to be computed.

You could also write a MATLAB function for plotting the array pattern. The plot should be in dB and give the result both as a function of wavenumber (*e.g.* as a function of k_x or $\sin(\text{angle})$), and angle. As default, plot only the visible region.

In addition you may also write a MATLAB function to analyze the array pattern, i.e. calculate the mainlobe widths, the maximum sidelobe level, the mean sidelobe level, etc.

Alternatively, you may use the provided functions – either as inspiration when writing your own, or as the functions to use. In the last case, make sure you understand what the functions do!

Grating lobes

4. Assume a linear array with $M = 24$ elements with unity weight. Let the elements be uniformly spaced and try four different element spacings: $d = \lambda/4, \lambda/2, \lambda$, and 2λ.
   Plot the results and discuss the differences as the inter-element distance varies.

Element weighting and element spacing

5. Consider the array with spacing $d = \lambda/2$ and $M = 24$. Apply several Kaiser windows to the array. Vary the β value from $0 - 5$ in steps of 0.5. How does the -3 dB or -6 dB mainlobe width and the maximum sidelobe level vary with β? Compute and report the white noise gain for each Kaiser window. What is the advantage and disadvantage of using for instant a Kaiser window on the array?

6. Let the array have $M = 24$ elements with unity weights and let the element spacings be given as in the variable *ElPos* given below:

```
% Array "Case B"
n = 1:12;
```

```
d = 1/2;   % Corresponding to lambda/2
e_n = [-0.017 -0.538 -0.617 -1.0 -1.142 -1.372 -1.487 ...
                     -1.555 -1.537 -1.3 -0.772 -0.242];
d_n = (n + e_n)*d;
ElPos = [-fliplr(d_n) d_n];
```

[The element locations have been taken from *Hodjat, F., and S. Hovanessian. "Nonuniformly spaced linear and planar array antennas for sidelobe reduction." IEEE Trans. on Antennas and Propagation 26.2 (1978): 198-204.*]

Compare the solutions from 4), 5) and 6) with respect to beamwidth, aperture, sidelobes, grating lobes, white noise gain, no. of elements, size of aperture, etc.

7. Apply steering to the array in 6) and plot the response. Also plot the unsteered array pattern for $u = \sin(\text{angle}) \in [-2,\ 2]$. Explain the behavior. Does the observed effect/feature also occur for the other arrays you have tested?

8. Compute and report the -3 dB and -6 dB mainlobe widths as a function of steering angle for the array in 6). Report the width both as k_x or $\sin(\text{angle})$), and angle. Comment on the result.

Thinning

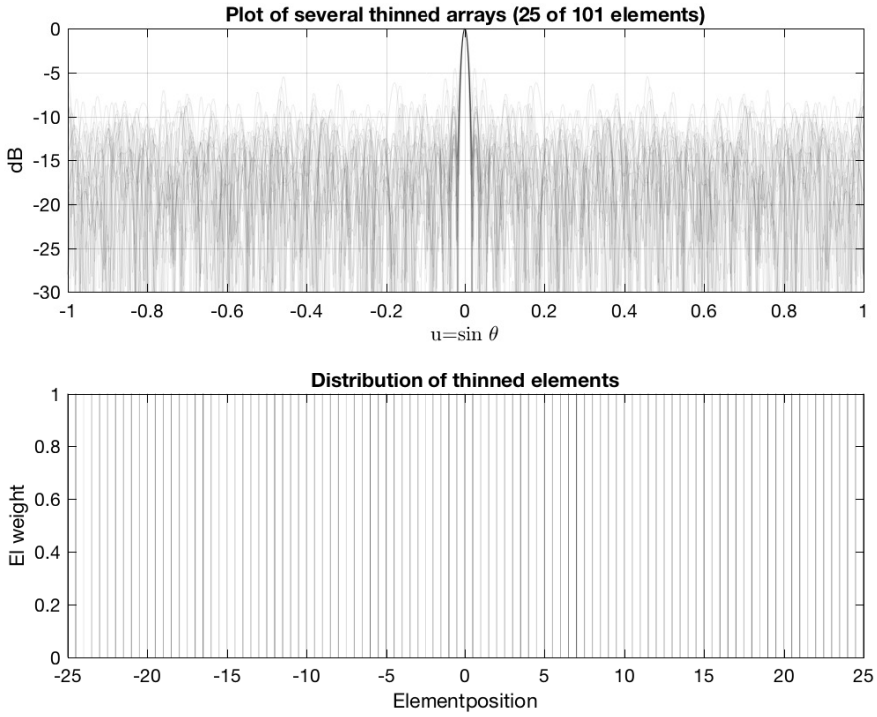
9. Assume an M=101 element array with $d = \lambda/2$. To make a thinned (or sparse) array with N=25 elements without limiting the aperture size (i.e. keeping the outer elements), and where the element positions are drawn from a uniform distribution, the following code can be used:

```
NEls = 101;
NPos = 25;
pos = [];
while length(pos)<NPos - 2
     pos = unique(ceil((NEls-2)*rand(1,NPos*2)),'stable');
end
ElPos = [-(NEls-1)/2  sort(pos(1:NPos-2))-(NEls-1)/2 (NEls-1)/2]*d
```

If the response of the thinned array above is stored in the variable *Resp*, it is possible to plot the response and the element positions for this array, and several other similar arrays, with the following code:

```
subplot(2,1,1)
plot1 = plot(u,db(abs(Resp)/max(abs(Resp))),'k'); grid on; ylim([-30 0]);
plot1.Color(4) = 0.075;
title(['Plot of several thinned arrays (', ...
    int2str(NPos),' of ',int2str(NEls),' elements)']);
ylabel('dB');
xlabel('u=sin $\theta$','interpreter','latex');
hold on
subplot(2,1,2)
plot2 = plot([ElPos; ElPos],[zeros(1,25); ones(1,25)],'k');
for ii=1:length(plot2)
plot2(ii).Color(4) = 0.075;
end
title(['Distribution of thinned elements']);
ylabel('El weight');
xlabel('Elementposition');
hold on
```

When plotting the response and the element position of 25 different thinned arrays on top of each other, the result looks as this:



The upper figure gives an indication of how the mainlobe width and the heights of the sidelobes varie. The lower figure indicates the variation in the element positions (which is uniform in this case).

Utilizing the code given above, estimate the mainlobe width (mean and standard deviation), the mean sidelobe level, and the maximum sidelobe level for larg collection of thinned array realizations. Compare your result to a dense array (all elements active) and one or a few of "the optimal" arrays found in *J. F. Hopperstad and S. Holm,"Optimization of sparse arrays by an improved simulated annealing algorithm", Proc. Int. Workshop on Sampling Theory and Applications, pp. 91-95, Loen, Norway, August 1999* available at https://www.researchgate.net/publication/228643183_Optimization_of_sparse_arrays_by_an_improved_simulated_annealing_algorithm.

Redo the analysis with the number of active elements equal to N=50 and N=75 and comment on how the mainlobe widths and sidelobe levels change when the number of active elements is increased.

10. Redo the analysis performed in 9), but let the element positions be drawn from a Normal distribution. You should try several values of the standard deviation (for instant 8 elements and 16 elements). How does this change the array performance?

    Hint: The following code just happens to produce a thinned array with normal distributed element layout (standard deviation = 16.83 elements):

```
% Normal dist positions ...
pos = [];  % Might be enough ...
while length(pos)<NPos-2
    tmp = ceil(NEls/6*randn(1,NPos*2));
    pos = unique(tmp(abs(tmp)<(NEls-1)/2),'stable');
end
ElPos = [-(NEls-1)/2  sort(pos(1:NPos-2))  (NEls-1)/2]*d
```

11. Again, redo the analysis performed in 9), but this time design a binned sparse array (as described in *Holm et. al, "Sampling theory and practice", Chapter 19* available at https://www.researchgate.net/publication/266499627_Sparse_Sampling_in_Array_Processing). How does this change the array performance?

Element directivity

12. Consider again the arrays in 4) (i.e. $M = 24$) with $d = \lambda$ and $d = 2\lambda$. Assume that each element is of size d and include the element response in the computation of the array pattern. Compare with the results in 4) and discuss the differences.

13. Apply steering to the array with the element response in the previous paragraph and see what happens as the steering angle is increased. Comment on your results.

14. Linear arrays with element spacing larger than $\lambda/2$ are regularly used in medical ultrasound imaging systems. Assume 24 unity weighted, regularly spaced elements and an element of size d equal to the element spacing. Design a linear array with as large aperture as possible that makes it possible to steer $\pm 15°$ with acceptable sidelobe and grating lobe levels.

The presentation

Please provide an electronic presentation with ideally one slide per question containing a short/minimal description of the problem, all necessary derivations, results, essential m-code (or python code), and discussions. In addition to the presentation, please also provide your matlab (or python) source code and a run-time example of you code (in Matlab, use 'diary', in an xterm, use for instant 'script').

Your presentation should not exceed 15-20 slides. The time for you to present it should be $\leq$15 minutes, i.e. prepare!!!

If you prefer to write some of the presentation or make some illustrations by hand, it is okay. Please then scan the handwritten pages and include these in your presentation.

Deliver your files (presentation, source code, and runtime example) in time in Devilry.

MATLAB code

MATLAB code is available from the lecture plan.